CO1 - Assessment Tool 2 - Application Based Questions

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Course code: DSA0503

Subject : Query Processing for Data Science

Experiment 1: Water Quality Monitoring (10 Marks)

Aim

To identify missing values, remove duplicate records, calculate the average Water Quality Index (WQI) for each location, and save the cleaned dataset.

Location	WQI	pH	Dissolved_Oxygen
Chennai	82	7.2	6.5
Chennai	85	7.3	6.8
Coimbatore	90	7.5	7.1
Madurai	76	7.0	6.2
Madurai	76	7.0	6.2
Salem	88	7.4	6.9
Chennai		7.1	6.4
Coimbatore	92		7.2
Salem	87	7.3	

Procedure

1. Import Pandas.
2. Load the water quality dataset.
3. Check missing values.
4. Remove duplicate records.
5. Fill missing values using the column mean.
6. Group data by location.

7. Calculate average WQI.
8. Save the cleaned dataset.
9. Display the results.

CODE:

```
import pandas as pd

df = pd.read_csv("water_quality.csv")

print("Original Dataset")

print(df)

print("\nMissing Values")

print(df.isnull().sum())

df = df.drop_duplicates()

df = df.fillna(df.select_dtypes(include='number').mean())

average_wqi = df.groupby("Location")["WQI"].mean()

print("\nAverage Water Quality Index")

print(average_wqi)

df.to_csv("cleaned_water_quality.csv", index=False)

print("\nCleaned dataset saved successfully.")
```

OUTPUT:

```
Missing Values
Location            0
WQI                 1
pH                  1
Dissolved_Oxygen    1
dtype: int64

Dataset after Removing Duplicates and Filling Missing Values
     Location        WQI        pH  Dissolved_Oxygen
0     Chennai  82.000000  7.200000          6.500000
1     Chennai  85.000000  7.300000          6.800000
2  Coimbatore  90.000000  7.500000          7.100000
3     Madurai  76.000000  7.000000          6.200000
5       Salem  88.000000  7.400000          6.900000
6     Chennai  85.714286  7.100000          6.400000
7  Coimbatore  92.000000  7.257143          7.200000
8       Salem  87.000000  7.300000          6.728571

Average Water Quality Index for Each Location
Location
Chennai       84.238095
Coimbatore    91.000000
Madurai       76.000000
Salem         87.500000
Name: WQI, dtype: float64

Cleaned dataset saved successfully as 'cleaned_water_quality.csv'
```

Experiment 2: University Alumni Database (10 Marks)

Aim

To integrate alumni information stored in a JSON file with employment details stored in a CSV file, identify alumni working in different industries, and generate a categorized alumni report.

Dataset 1 (alumni.json)

```
[
 {
  "ID":1,
  "Name":"Arun",
  "Department":"CSE"
 },
 {
  "ID":2,
  "Name":"Priya",
  "Department":"ECE"
 },
 {
  "ID":3,
  "Name":"Rahul",
  "Department":"IT"
 },
 {
  "ID":4,
  "Name":"Divya",
  "Department":"EEE"
 }
]
```

Dataset 2 (employment.csv)

ID	Company	Industry
1	Infosys	IT
2	Apollo Hospital	Healthcare
3	TCS	IT
4	L&T	Engineering

Procedure

1. Import the Pandas library.
2. Load the alumni data from the JSON file.
3. Load the employment data from the CSV file.
4. Merge the datasets using the common **ID** field.
5. Display the integrated dataset.
6. Group alumni by **Industry**.
7. Generate the categorized alumni report.
8. Save the final report as a CSV file.
9. Display the categorized report and confirm that it has been saved successfully.

CODE:

```
import pandas as pd
alumni = pd.read_json("alumni.json")
employment = pd.read_csv("employment.csv")
merged = pd.merge(alumni, employment, on="ID")
print("Integrated Alumni Dataset")
print(merged)
industry_report = merged.groupby("Industry")["Name"].apply(list)
print("\nCategorized Alumni Report")
print(industry_report)
merged.to_csv("alumni_report.csv", index=False)
```

print("\nReport saved successfully.")

OUTPUT:

```
Alumni Dataset
   ID     Name Department
0   1     Arun        CSE
1   2    Priya        ECE
2   3    Rahul         IT
3   4    Divya        EEE
4   5  Karthik       MECH

Employment Dataset
   ID          Company       Industry
0   1          Infosys             IT
1   2  Apollo Hospital     Healthcare
2   3              TCS             IT
3   4              L&T    Engineering
4   5    Ashok Leyland  Manufacturing

Integrated Alumni Dataset
   ID     Name Department          Company       Industry
0   1     Arun        CSE          Infosys             IT
1   2    Priya        ECE  Apollo Hospital     Healthcare
2   3    Rahul         IT              TCS             IT
3   4    Divya        EEE              L&T    Engineering
4   5  Karthik       MECH    Ashok Leyland  Manufacturing

Categorized Alumni Report
Industry
Engineering            [Divya]
Healthcare             [Priya]
IT               [Arun, Rahul]
Manufacturing        [Karthik]
Name: Name, dtype: object
```